18.745 Lie Algebra

Homework 2

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Problem 1 — Let V be an $\mathbb{C}$-vector space of dimension n equipped with a nondegenerate skew-symmetric bilinear form $(\bullet, \bullet)$.

1. Prove that $n = 2m$ is even.
2. There is a basis $\{v_1, \ldots, v_m, w_{-m}, \ldots, w_{-1}\}$ of V satisfying the following: for any $x, y \in V$ with $x = \sum_{i=1}^{m} a_i v_i + \sum_{j=-m}^{-1} b_j w_j$ and $y = \sum_{i=1}^{m} a_i' v_i + \sum_{j=-m}^{-1} b_j' w_j$,
$$(x, y) = \sum_{i=1}^{m} a_i b_i' - \sum_{j=1}^{m} a_j' b_j.$$
3. Let $Sp(2m, \mathbb{C})$ be the group of isometries of $(\bullet, \bullet)$. Compute the dimension of $Sp(2m, \mathbb{C})$ over $\mathbb{C}$.

Any bilinear form can be writen as a matrix A such that $(x, y) = x^\top A y$. We observe that $A = -A^\top$. Because the multiset of eigenvalues are preserved under transpose, we find that A must have eigenvalues that come in pairs $(\lambda, -\lambda)$ (after all, there cannot be any 0's, else the form would be degenerate).

For the second part, we induct. When $m = 0$, there is nothing to prove. Suppose the statement is true for $m - 1$. Let V be a vector space of dimension $2m$ equipped with a nondegenerate skew-symmetric bilinear form $(\bullet, \bullet)$. Pick any $v_1 \in V$. By non-degeneracy, there exists a v_2 such that $(v_1, v_2) \neq 0$, which we will fix such that $(v_1, v_2) = 1$. Fill out the remaining basis $v_3, \cdots v_n$ arbitrarily. We will now perform many basis changes. For all v_i, $i > 2$, let
$$v_i' = v_i - \frac{(v_1, v_i)}{(v_1, v_2)} v_2 \implies (v_1, v_i') = 0.$$
Then, let
$$u_i = v_i' - \frac{(v_2, v_i')}{(v_2, v_1)} v_1 \implies (v_1, u_i) = (v_2, u_i) = 0.$$
Finally, let $u_1 = v_1/(v_1, v_2)$ and $u_2 = v_2$ such that:
$$(u_1, u_i) = \begin{cases} 1 & i = 2 \\ 0 & i \neq 2 \end{cases} \qquad \text{and} \qquad (u_2, u_i) = \begin{cases} 1 & i = 1 \\ 0 & i \neq 1 \end{cases}.$$
At this point, $Span(u_3, \cdots u_n)$ is a subspace of V of dimension $2m - 2$ equipped with a nondegenerate skew-symmetric bilinear form $(\bullet, \bullet)$ of its own right. By induction, we

find a basis $w_1, w_{-1}, \cdots w_{-m}, w_m$ such that

$$(w_i, w_j) = \begin{cases} 1 & i = -j > 0 \\ -1 & i = -j < 0 \\ 0 & i \neq -j \end{cases}.$$

Then, the problem statement follows by linearity.

For the third part, observe that choices of $M \in Sp(2m, \mathbb{C}) \subset \mathbb{GL}_{2m}(\mathbb{C})$ are equivalent to choices of basis; the question is really asking for the number of ways to pick a basis satisfying the above conditions.

We claim the answer of $m(2m+1)$; we repeat the steps of the above proof.

1. There are $2m$ dimensions to select the first vector v_1 (you can pick anything in $\mathbb{R}^{2m}$).

2. There are $2m-1$ dimensions to select v_2 such that $(v_1, v_2) = 1$. This is clear from the final result; $v_2 = v_1 + \sum_{\beta_i} v_i$ where β_i are the coefficients for $i \in \{-m, \cdots -2, 1, \ldots m\}$.

Thus,

$$(m-1)(2(m-1)+1) + (2m-1) + 2m = m(2m+1)$$

as desired.

Problem 2 — Let X be a real manifold. Show that a vector bundle E of rank n on X with $p : E \to X$ is trivial if and only if it admits sections $s_1, \ldots, s_n$ which form a basis in every fibre $p^{-1}(x)$.

This is pretty stupid. If E is trivial, then there exists a diffeomorphism $\phi : E \to X \times \mathbb{R}^n$ such that $p = \pi_X \circ \phi$ where π_X is the projection $X \times \mathbb{R}^n \to X$. Then, the sections s_i are given by $s_i(x) = \phi^{-1}(x, e_i)$ where e_i is the ith standard basis vector in $\mathbb{R}^n$.

On the other hand, if you have such a collection of sections, then you can define a map $\phi : E \to X \times \mathbb{R}^n$ by $\phi(x, \sum_i a_i s_i(x)) = (x, a_1, \ldots a_n)$. This is clearly a bijection and is smooth since the s_i are smooth. Thus, E is trivial.

Problem 3 —

1. Consider $\mathbb{SL}(2,\mathbb{R})$. Show that the image of the exponential mapping for $\mathbb{SL}(2,\mathbb{R})$ consists of precisely those matrices $A \in \mathbb{SL}(2,\mathbb{R})$ with $\mathrm{Tr}(A) > -2$, together with $-I$. Show that the image of the exponential map is not dense in $\mathbb{SL}(2,\mathbb{R})$.

2. Suppose G is a connected, commutative closed Lie subgroup of $\mathbb{GL}_n$. Show that the exponential map $\exp : \mathfrak{g} \to G$ is surjective.

3. The exponential mapping $\exp : \mathfrak{u}(n) \to \mathbb{U}(n)$ is onto, but not one to one. (Hint: A unitary matrix has an orthonormal basis of eigenvectors)

(Part 1)

Recall that the exponential mapping on $\mathbb{SL}(2,\mathbb{R})$ coincides with the usual matrix exponent. In particular, we will make extensive use of

$$Be^A B^{-1} = e^{BAB^{-1}}.$$

for all $B \in \mathbb{GL}(2,\mathbb{R})$.

As such, first observe that for some generic $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\exp \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1+a & b \\ c & 1+d \end{pmatrix} + O(a^2, b^2, c^2, d^2, ab, ac, ad, bc, bd, cd).$$

thus the desired condition is that the trace is 0 to first order; this agrees with the fact that $\mathbb{SL}(2,\mathbb{R})$ is dimension 3. The eigenvalues are thus $(\lambda, -\lambda)$ for $\lambda \in \mathbb{C}$. Furthermore, the determinant of A is pure real, thus λ is either pure imaginary or pure real.

Under suitable basis change B a la Jordan Canonical Form (over $\mathbb{C}$), any element of $\mathbb{SL}(2,\mathbb{R})$ is thus:

- For all $\lambda \in \mathbb{R} \cup i\mathbb{R}$,
$$BAB^{-1} = \begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix}$$

- If $\lambda = 0$, there is an additional possibility:
$$BAB^{-1} = \begin{pmatrix} 0 & \kappa \\ 0 & 0 \end{pmatrix}$$

Okay, great. These exponentiate to:

- $$Be^A B^{-1} = \begin{pmatrix} e^{\lambda} & 0 \\ 0 & e^{-\lambda} \end{pmatrix}$$

  This yields all e^A which have two eigenvectors, whose eigenvalues are of the form $(e^{\lambda}, e^{-\lambda})$ for $\lambda \in \mathbb{R}$, or $(e^{i\lambda}, e^{-i\lambda})$ for $\lambda \in \mathbb{R}$. This comprises a subset of the *diagonalizable* matrices in $\mathbb{SL}(2,\mathbb{R})$; notably exempt are the matrices whose eigenvalues are of the form $(-e^{\lambda}, -e^{-\lambda})$.

  So this case captures all diagonalizable matrices with $\mathrm{Tr}\, e^A > -2$, as well as $-I$.

- If we are in the nilpotent case, then

$$Be^A B^{-1} = \begin{pmatrix} 1 & \kappa \\ 0 & 1 \end{pmatrix}$$

  These are exactly all non-diagonalizable matrices in $\mathbb{SL}(2, \mathbb{R})$, completing the characterization in the problem.

Cool. We now show that this image is not dense in $\mathbb{SL}(2, \mathbb{R})$. This is basically obvious; consider the matrix

$$\begin{pmatrix} -2, & 0 \\ 0, & -\frac{1}{2} \end{pmatrix}$$

All matrices in a neighborhood of this matrix will have eigenvalues close to -2 and $-\frac{1}{2}$, thus the trace will be close to $-\frac{5}{2}$, which is not in the image of the exponential map.

(Part 2)

The exponential map is a local diffeomorphism from a neighborhood $\mathfrak{u} \subset \mathfrak{g}$ of 0 onto a neighborhood of $U \subset G$ of 1. This neighborhood generates G, because G is connected. Thus, all elements of G can be written

$$g = \exp(X_1) \cdots \exp(X_n)$$

for some $X_i \in \mathfrak{u}$. However, since G is commutative,

$$g = \exp(X_1 + \cdots + X_n)$$

and we are done.

(Part 3)

Uh, it's not that deep. $\mathfrak{u}(n) \subset \mathfrak{gl}_n$ is a real vector space. By staring,

$$\exp(x) \approx 1 + x$$

thus we must have

$$1 = (1 + x)(1 + x)^\dagger \approx 1 + x + x^\dagger$$

thus $x = -x^\dagger$ i.e. $\mathfrak{u}(n)$ consists of precisely the anti-Hermitian matrices.

Why is this onto? Well, using the basis-changing strategy from before, any anti-Hermitian matrix is of the form

$$\begin{pmatrix} ia_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & ia_n \end{pmatrix}$$

where $a_i \in \mathbb{R}$. Then, these exponentiate to

$$\begin{pmatrix} e^{ia_1} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & e^{ia_n} \end{pmatrix}$$

Thus, by basis changing, one can create any matrix with eigenvalues of the form e^{ia_i}; these are precisely the unitary matrices.

On the other hand, clearly $\exp(0) = \exp(2\pi i \cdot I) = I$, so the exponential map is not injective.

Problem 4 — Recall from the first pset, there is a homomorphism $\phi : \mathbb{SU}(2) \to \mathbb{SO}(3)$. Compute the Lie algebra homomorphism $d\phi : \mathfrak{su}(2) \to \mathfrak{so}(3)$ explicitly.

LOL wait just a moment. ϕ is just $\mathrm{Ad} : \mathbb{SU}_2 \to \mathbb{GL}(\mathfrak{su}_2)$ where $\mathfrak{su}_2 \equiv \mathbb{R}^3$. So if we take an orthonormal basis of $\mathfrak{su}_2$ called

$$\sigma_1 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad \sigma_2 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \sigma_3 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

this ends up as a subgroup of $\mathbb{GL}(\mathbb{R}^3)$... which happened to end up being $\mathbb{SO}(3)$. (This makes sense, because as a an inner product space the norm is $\mathrm{Tr}(AB)$, but this trace is preserved under conjugating A and B. It's not deep.)

So now, you want me to compute $d\,\mathrm{Ad} : \mathfrak{su}_2 \to \mathfrak{gl}(\mathfrak{su}_2)$, which is going to map into $\mathfrak{so}_3$. But we know that $d\,\mathrm{Ad}$ sends x to $[x, \bullet]$ from class. So I guess I can just compute it...

$$d\,\mathrm{Ad}(\sigma_1) = [\sigma_1, \bullet] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$
$$d\,\mathrm{Ad}(\sigma_2) = [\sigma_2, \bullet] = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$
$$d\,\mathrm{Ad}(\sigma_3) = [\sigma_3, \bullet] = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

So... yeah. That's it.

Problem 5 — Here, $K = \mathbb{R}$ or $\mathbb{C}$ and the Lie groups and Lie algebras are defined over K.

1. If $\phi : \mathfrak{g}_1 \to \mathfrak{g}_2$ is a K-Lie algebra homomorphism, show that $\ker \phi$ is an ideal in $\mathfrak{g}_1$.
2. Find the kernel of $\mathrm{ad} : \mathfrak{gl}_n \to \mathfrak{gl}(\mathfrak{gl}_n)$.
3. Prove that $\mathfrak{gl}_n \cong \ker(\mathrm{ad}) \oplus \mathfrak{sl}_n$.
4. Find the Lie algebra of the group $PGL_n = \mathbb{GL}_n/Z$ where Z is the center of $\mathbb{GL}_n$.

(Part 1)

Let $x \in \ker \phi$ and $y \in \mathfrak{g}_1$. Then, $\phi([x, y]) = [\phi(x), \phi(y)] = 0$, so $[x, y] \in \ker \phi$.

(Part 2)

ad is just the map $x \mapsto [x, \bullet]$. Thus, $\ker \mathrm{ad}$ is the set of elements x such that $[x, y] = 0$ for all y. This is precisely the center of $\mathfrak{gl}_n$, which coincides with the word "center" in the ring-theoretic sense. Well okay fine; if x commutes with the matrix with a 1 at (i, i) and 0's everywhere else, we find that the x has 0's on all off-diagonal entries. Then, if x commutes with a matrix with a 1 at (i, j) and 0's everywhere else, we find that $x_{i,i} = x_{j,j}$. Thus, x must be a multiple of the identity; these are all clearly part of the center.

(Part 3)

I mean, $\mathfrak{sl}_n$ consists of precisely the matrices with trace 0. $\mathfrak{gl}_n$ consists of all matrices. Thus, $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{R}I$, as desired.

(Part 4)

Z consists of constant multiples of the identity; this can be quickly seen from analysis similar to part 2. By the correspondence between the center of a Lie group and its Lie algebra, I know that Z consists of precisely those matrices of the form $\alpha \cdot I$ for $\alpha \in \mathbb{R}_{\geq 1}$. In the lie algebra case, this corresponds to modding out by $\mathbb{R} \cdot I$, which yields $\mathfrak{sl}_n$.